Team possession from automatic football tracking

V5 technical report

Barca4Ever

More Than A Hack 2027 – Initial Challenge
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Abstract

We estimate team possession from player and ball tracks with missing observations and fragmented player identities. V5 combines a geometry classifier with a trajectory network, then uses learned directed transitions to decode possession spells. The final model was fitted on four licensed external matches and selected using a fifth. In reciprocal cross-validation within the four fitting matches, spellF1 was 0.7133 on clean tracking and 0.5344 with simulated noise. The collection has been inspected repeatedly during development, so these are development results with limited independent evidence. Noisy short possessions remain the main weakness. Our pipeline runs offline on a CPU and produces one label for each of the 199,319 tactical-video frames.

1 Task and evaluation

The output is a sequence of home possession (H), away possession (A), dead ball (D), and abstention (X). A useful estimate must recover when possessions start and end. Frame accuracy can hide errors on brief possessions, while rapidly alternating labels can create possessions that never occurred.

The organizer ranks submissions by `spellF1`, with frame-level `macroF1` breaking ties. A possession spell is a maximal uninterrupted H or A interval inside a scored reference span. Predicted and reference spells are matched one-to-one within the same team by temporal intersection-over-union (IoU). F1 is computed at IoU thresholds $0.3, 0.4, 0.5, 0.6, 0.7$ and averaged. Dead-ball intervals are excluded from the spell list. Unknown reference spans are excluded. Predicted X values are forward-filled, with leading abstentions back-filled inside each valid span. The separate short-recovery count uses spells shorter than two seconds and IoU 0.5. These counts are distinct from the multi-threshold headline score.

V5 takes tactical player and ball XY positions and team assignments. We chose to work from these tracks. Video, Smart Tagging, broadcast tracking and homography are outside this implementation. All fitting, selection and quality evaluation use external labels. The supplied match is used for local inference.

2 Method

Figure 1 shows the inference path. Geometry supplies direct spatial evidence. The trajectory network adds local motion and observation history. The transition model helps the decoder place changes without accepting every brief fluctuation in the frame probabilities.

2.1 Input representation

The source tracks are sampled at 25 Hz. The loader maps source record 1 to output frame 0, converts normalized positions to pitch-centred metres, and checks record order and slot counts. Slots are track fragments rather than stable player identities. The network retains all observed actors in padded team groups and masks missing actors. Goalkeeper-role inputs stay zero.

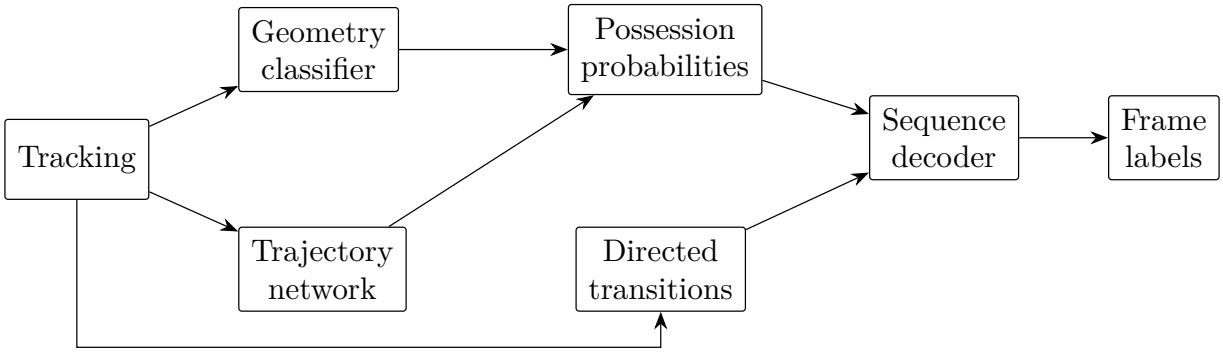


Figure 1: V5 inference. The trajectory branch is inspired by contact modeling but is trained on possession and ball-status labels, without touch-level supervision.

Ball gaps of up to two seconds are linearly interpolated between observed endpoints. Longer gaps remain missing. Raw visibility and time since/until a ball observation accompany the filled coordinates, so interpolation does not erase the original missingness. A complete player-tracking outage of at least ten seconds receives X and separates inference blocks. Missing ball observations alone do not assign D.

2.2 Possession probabilities

The geometry branch is a histogram gradient boosting classifier over 59 spatial, motion and visibility features. The neural branch adapts the shared trajectory temporal convolutional network (TCN) and ball-query attention approach of Borghesi et al. [1]. We implemented and trained this branch ourselves. The authors' original weights and confidential training data were unavailable.

The network predicts dead probability d_t and conditional home probability h_t. Its joint probabilities are

$$q_t = \big((1-d_t)h_t,\ (1-d_t)(1-h_t),\ d_t\big), \qquad p_t = \tfrac{1}{2}g_t + \tfrac{1}{2}q_t, \tag{1}$$

where g_t is the geometry classifier output in H, A, D order. The home loss is masked on dead-ball labels.

The TCN uses width 64 and six causal blocks with kernel size 2 and dilations $1, 2, 4, 8, 16, 32$. A target-frame multilayer perceptron (MLP) and distance-biased attention add local geometry. There are 20 input channels, including raw visibility and observation gap timing. Inputs are sampled at 16 Hz using actual elapsed source time for velocity. A window contains 64 samples with zero-based target index 47. It therefore includes about one second of future trajectories. Gap features can use up to ten seconds of future observation evidence. This is an offline method.

2.3 Changes of possession

A second boosting model estimates the six ordered changes between H, A, D, plus background. It uses 293 geometry/context features, with before, after and signed differences at offsets $\pm 2, 6, 12$ source frames. Training gives transition examples extra weight, so its scores are uncalibrated evidence.

For a sequence of T frames, the decoder selects

$$\hat{y} = \underset{y \in \{H,A,D\}^T}{\arg\max} \left[\sum_{t=1}^{T} \log p_t(y_t) - \sum_{t=2}^{T} \mathbf{1}[y_t \neq y_{t-1}] \frac{16}{1 + 8b_t(y_{t-1}, y_t)} \right], \tag{2}$$

where $b_t(i, j)$ is the directed transition score. A plausible H-to-A change reduces that switch cost specifically. The decoder is run separately in each tracking block.

3 Experiments and results

3.1 External data and selection

We use IDSSE, the public DFL/Sportec dataset described by Bassek et al. [2, 3]. The reference labels are the provider's `BallPossession` and `BallStatus` fields. Inactive ball status maps to D, live possession 1/2 to H/A, and unknown to X. Labels are isolated from features. The data are licensed under CC BY 4.0. Our changes include parsing, feature construction, simulated corruption and model training.

Role	Matches
Final fitting	J03WOH, J03WOY, J03WPY, J03WQQ
Selection	J03WR9
Reused regression	J03WMX, J03WN1

Table 1: Roles of the seven external matches. Both halves stay with their match.

The deployed model uses the four fitting matches. Cross-validation within those four fits WOH/-WOY and evaluates WPY/WQQ, then reverses. Each fold excludes its evaluated matches from every learned component, including the neural initializer. The two regression matches were originally held out, then reused during development. All seven matches have now been inspected repeatedly. Fitting exclusion within a fold does not make this collection an independent final test.

We selected input rate and mixture weights on WR9. The composite had the highest WR9 spellF1 among candidates that passed their predeclared clean/noisy cross-validation and regression acceptance checks. The selection protocol records those comparisons and tie-breaking rules. Its neural branch was fine-tuned for two epochs of 100,000 fresh references, sampled 75% by eligible frames and 25% by team spells. Detailed configurations, training seeds, checkpoint lineage and rejected alternatives are recorded in `V5_MODEL_GUIDE.md` and the `MULTIAGENT_RESULTS.md`.

3.2 Matched comparisons

V4.1 is the preceding geometry/trajectory pipeline with destination-based boundary scores. V5 adds raw observation-confidence channels, elapsed-time velocity correction, fresh training references and directed transitions. The table compares complete pipelines rather than isolating each change.

We use the same labels and corruption settings for each paired comparison. Synthetic noise adds 0.25 m player jitter, 0.35 m ball jitter, 5% player dropout and contiguous ball gaps of 15–84 source frames. Scores below average halves equally, while short-spell hits are pooled.

		spellF1		Short hits / total	
Comparison	View	V4.1	V5	V4.1	V5
Cross-validation	Clean	0.7105	0.7133	191/463	206/463
Cross-validation	Noisy	0.5329	0.5344	79/463	80/463
Reused regression	Clean	0.6808	0.6847	128/362	132/362
Reused regression	Noisy	0.4968	0.5140	64/362	71/362

Table 2: External development results. Short hits use duration below two seconds and IoU 0.5; spellF1 averages the five thresholds defined in Section 1.

The noisy cross-validation gain is 0.0015, with one additional short possession recovered. Noisy reused regression shows a larger difference. Repeated use of all seven matches limits the independent evidence these comparisons provide.

4 Limitations and reproduction

V5 recovered 80 of 463 short possessions in noisy cross-validation. Brief turnovers remain difficult, especially when ball observations are missing or actor trajectories are fragmented. The corruption model does not cover every real tracking failure. XY positions also leave whistles, aerial contact and body orientation ambiguous. Exact nearest-player ties retain the trained slot-order convention. Finally, one provider and seven repeatedly inspected matches give limited evidence about transfer to the competition match.

The repository includes the trained artifact and pinned Python 3.12 CPU dependencies. From its root, a POSIX shell can regenerate the file with

```
python -m pip install -r requirements-contact.txt
python refined_infer.py \
  --model models/external_v5.joblib \
  --data-dir /local/bundle \
  --out local_outputs/Barca4Ever.csv
python evaluation/validate_submission.py local_outputs/Barca4Ever.csv
```

The bundle contains `tactical/BAR-ATH_tactical_atd.xml` and its `.txt` tracking file. Windows users can run `Run-Barca4Ever.ps1`. CPU inference uses four threads. The output manifest records model and CSV hashes.

The November 11 release passed 85 synthetic tests and a 199,319-frame generated match through the launcher, V5 inference and the organizer's format validator. Earlier evidence records four external CPU parity comparisons and the original 82-test suite. These are software checks. The possession-quality evidence is Table 2. The competition score awaits the organizer's private evaluation.

Challenge inputs, predictions and local logs are excluded from the public package. Inference performs no fitting or calibration, and predictions were not checked against the supplied video or ground truth. The organizers prohibit uploading their match data or extracts to hosted AI services. Dataset attribution must accompany the trained artifacts.

Repository https://github.com/aqaPayam/Barca4Ever

References

[1] M. Borghesi, L. D. Costa, L. Morra and F. Lamberti. Using Temporal Convolutional Networks to estimate ball possession in soccer games. *Expert Systems with Applications*, 223, 119780, 2023. doi:10.1016/j.eswa.2023.119780.

[2] M. Bassek, R. Rein, H. Weber and D. Memmert. An integrated dataset of spatiotemporal and event data in elite soccer. *Scientific Data*, 12, 195, 2025. doi:10.1038/s41597-025-04505-y.

[3] IDSSE dataset. Original data from DFL / Sportec Solutions. doi:10.6084/m9.figshare.28196177. Licensed under CC BY 4.0.